



**PERTANDINGAN KEMAHIRAN
KOLEJ VOKASIONAL
PERINGKAT KEBANGSAAN KALI KE 10
(KV Skills 2026)**

**BIDANG KEMAHIRAN: WEB TECHNOLOGY
(6 JAM)**

MODUL C

SOALAN PRAKTIKAL

NAMA: _____

NO. KP: _____

**BAHAGIAN PENDIDIKAN DAN LATIHAN TEKNIKAL VOKASIONAL (BPLTV)
KEMENTERIAN PENDIDIKAN MALAYSIA**

Soalan ini mengandungi (12) halaman bercetak

MODULE C (6 HOURS)

CONTENTS

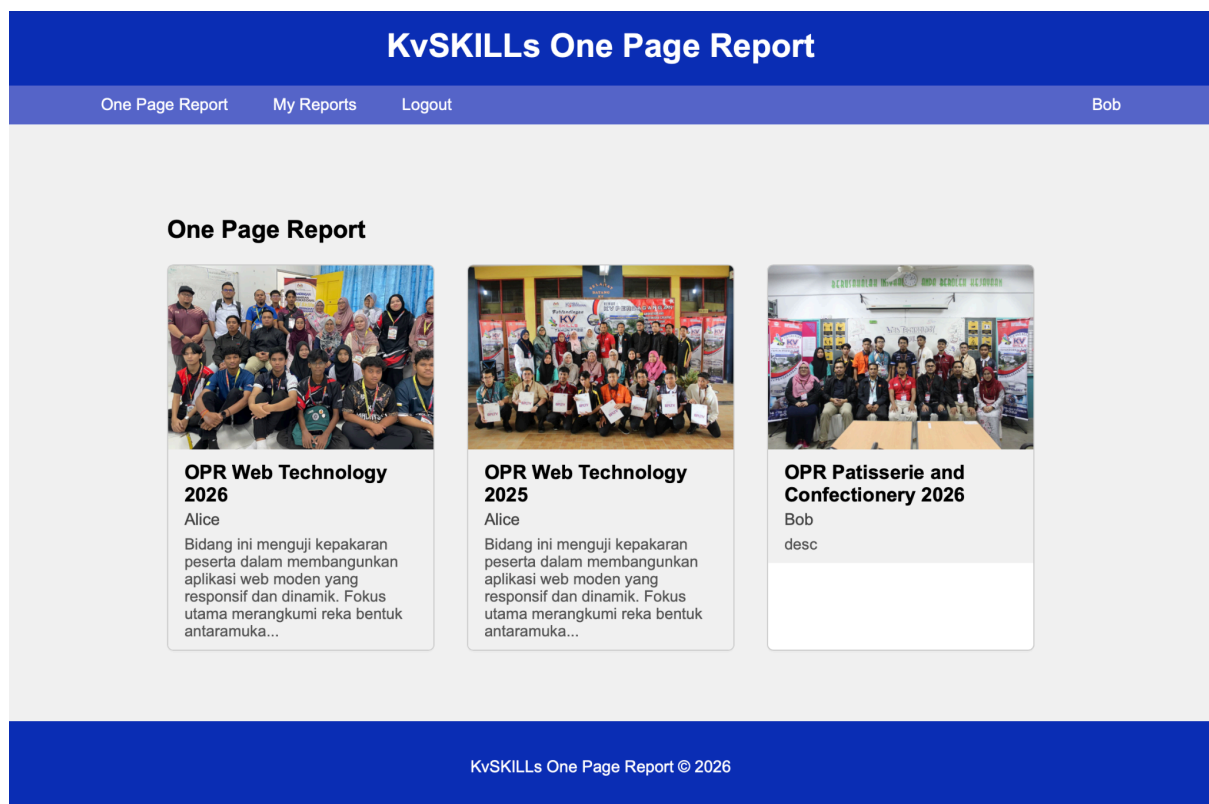
This module has the following files:

Module_C_Media_Files.zip

INTRODUCTION

You are to develop a **server-side (backend)** web application named **KvSkills One Page Report** that allows registered users to create one page report for an event, upload photos of event and make them available for other users to view.

This module tests your web development skills in **PHP 8, PHP Framework (Laravel 12)** and **MySQL (run under XAMPP setup)**. To aid your web development, jQuery are provided (but optional).



Your solution needs to be a PHP server-side project with the use of PHP Framework (Laravel 12)

DESCRIPTION OF PROJECT AND TASKS

In general, you are to develop the following pages or screens:

- **Index | Login and Logout** page(s)
- **Register** page
- **One Page Report** and **Report** pages
- **My Reports** and **Manage Report** pages
- **Users** page (Admin only)

Add additional pages files if necessary. Detailed requirements are described below.

1. Page Design and Layout

This module focuses on **backend web programming**. Little marks are awarded for page design and layout. You are recommended to keep your design simple, clean and neat. Generally, your design should fulfil the following requirements:

- Layout: Contain distinct header, navigation, main and footer sections.
- User Experience: Smooth, easy-to-use and user-friendly.

NOTE: Figures used below are to aid the explanations. It is NOT necessary to follow the screen designs exactly. You can have your own page design and layout, as long as the requirements are met.

2. Database

You are required to store the following records in the **MySQL** database:

2.1 User Records

Generally, a user record contains the following fields:

- **Id** – Primary key, int(20).
- **Username** – Unique. For login purpose.
- **Password** – Password (hashed). For login purpose.
- **Role** – 0 for user. 1 for admin.
- **Category_id** – Foreign key referred from **Categories** table.

Pre-insert the following user record (admin) for testing purpose:

Id	Username	Password	Role	Category_id
1	Admin	Password (hashed)	1	
2	Alice	Password (hashed)	0	
3	Bob	Password (hashed)	0	

2.2 Reports Records

Generally, an albums record contains the following fields:

- **Id** – Primary key, int(20).
- **User_id** - Foreign key, int(20).
- **Category_id** - Foreign key, int(20).
- **Title** – varchar(255).
- **Description** - varchar(255).
- **Date** - date.

2.3 Photos Records

Generally, a photos record contains the following fields:

- **Id** – Primary key, int(20).
- **Report_id** - Foreign key, int(20).
- **Path** – path for photo uploaded, varchar(255).

2.4 Categories Records

Generally, a likes record contains the following fields:

- **Id** – Primary key, int(20).
- **Category_name** - varchar(255).

3. Index | Login Page

The image shows a login form with a title 'Index | Login' in blue. Below the title, there are two text labels, 'Username :' and 'Password :', each followed by a white rectangular input box. At the bottom of the form, there are two buttons: a blue button labeled 'Login' and a purple button labeled 'Reset'.

Figure C2 : Index | Login page

Create the Login page (Figure C2). Requirements:

3.1 General

Basic input validations: All fields are required.

Login button: Login the user based on records in the database.

- **If user role is 0 (user). Redirect to Gallery page.**

- **If user role is 1 (admin). Redirect to Users page.**

Reset button: Reset the form.

Show appropriate error message if username and password not matched.

3.2 Navigation Menu

Navigation menu should be updated accordingly based on user login status.

Show the username on top of the page after login (Figure C3, C4 and C5).



Figure C3 : Before Login



Figure C4 : After login as user



Figure C5 : After login as admin

4. Logout Page

Create the **Logout** page. Requirements:

- The page DOES NOT have user interface (contains only logout logic).
- Logout the user. Redirect to Index | Login page.

5. Register Page

Create the **Register** page (**Figure C6**). Requirements:

Register

Username :

Password :

Confirm Password :

Kategori :

Figure C6 : **Register** page

- Basic input validations: All fields are required. Password and confirm password must matched.
- Category dropdown input are loaded from Categoris table.
- Register button: Register the user. Redirect to Index | Login page.
- Reset button: Reset the form.
- Show appropriate error message if username duplicated.

6. One Page Report Page

Create the **One Page Report** page (**Figure C7**). Requirements:

One Page Report

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Alice

Bidang ini menguji kepakaran peserta dalam membangunkan aplikasi web moden yang responsif dan dinamik. Fokus utama merangkumi reka bentuk antaramuka...

OPR Web Technology 2025

Alice

Bidang ini menguji kepakaran peserta dalam membangunkan aplikasi web moden yang responsif dan dinamik. Fokus utama merangkumi reka bentuk antaramuka...

OPR Patisserie and Confectionery 2026

Bob

Bidang ini mempamerkan gabungan unik antara sains kulineri dan kreativiti artistik dalam penghasilan produk pencuci mulut bertaraf tinggi. Peserta perlu...

Figure C7 : **One Page Reports** page

- List all public and non-empty albums of all users.
- For each album:
 - Use the first photo as its thumbnail (visually cropped and resized to fit in square box).
 - Show album's title and owner's username.
- Click on thumbnail: Redirect to Slide page. Show respective photos of the selected album.

7. Report Page

Create the Slide page (Figure C8). Requirements:

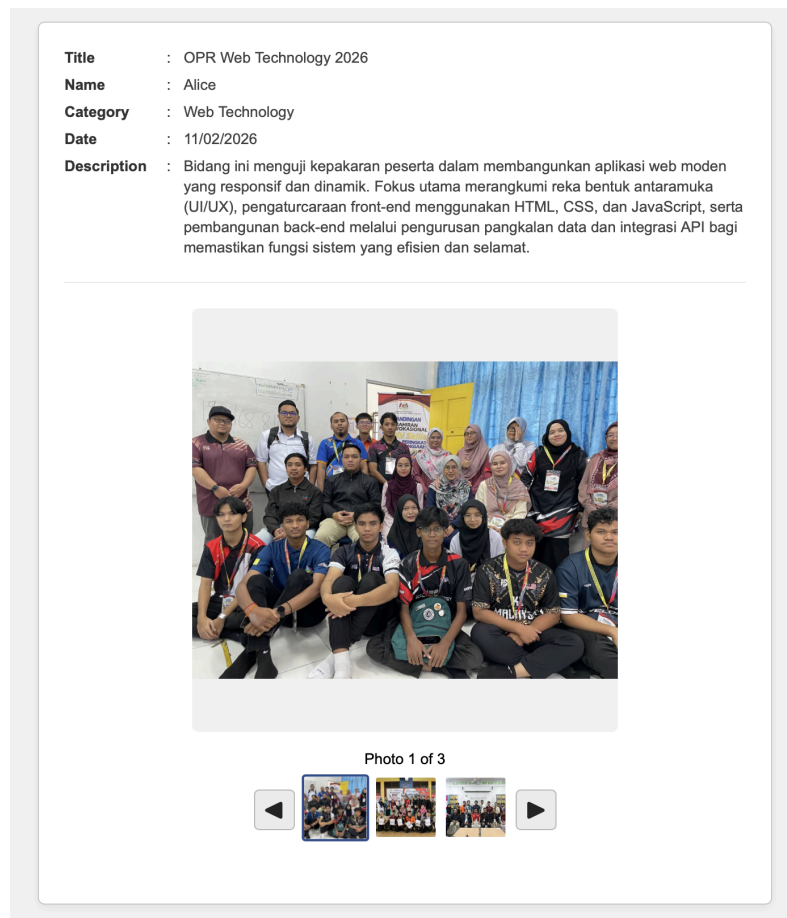


Figure C8 : Slide page

- Top section: Show album's title, owner's username, category which is belong to user, date and description.
- Middle section: Show the selected photo in its original size and aspect ratio. By default, the first photo.
- Bottom section:
 - List all photos in the selected album as thumbnails (visually cropped and resized to fit in square box).
 - Click on thumbnail: Select the photo. Show the original photo in the middle

section.

- o Previous button: Select the previous photo.
- o Next button: Select the next photo.
- o Selected photo should be highlighted. Update photo number and count accordingly.
- o Use JavaScript/AJAX techniques for this to prevent page reload.

8. MyReports Page

Create the **MyReports** page (Figure D10). Requirements:

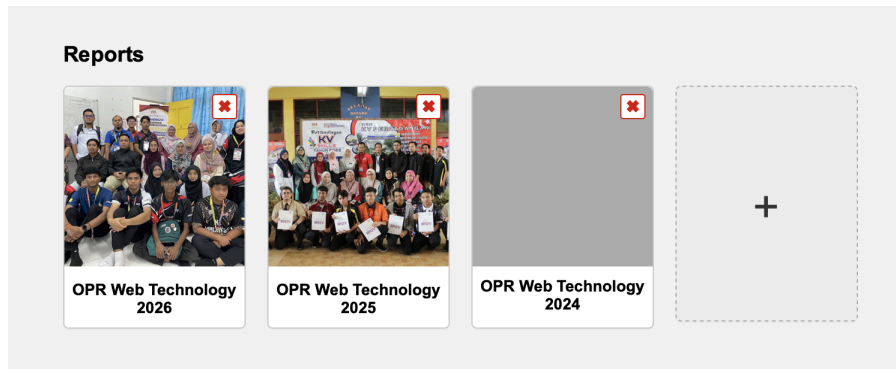


Figure C10 : **MyReports** page

- List all reports of the user.
- For each report:
 - o Use the first photo as its thumbnail (visually cropped and resized to fit in square box).
 - o If the report is empty, fill the square box with solid color.
 - o Show report's title.
 - o Delete (cross) button: Delete the album and its photos.
- Click on thumbnail: Redirect to Manage Report page. Show respective photos of the selected album.
- Add (Plus) button: Show a modal dialog to add report (Figure C11).
 - o Basic input validations: Title is required, Description is required, Date is required. Status is either public (0) or private (1).
 - o Category are display based on Category_Id
 - o Add button: Create the report. Redirect to Manage Report page.
 - o Cancel button: Reset the form. Close modal dialog.

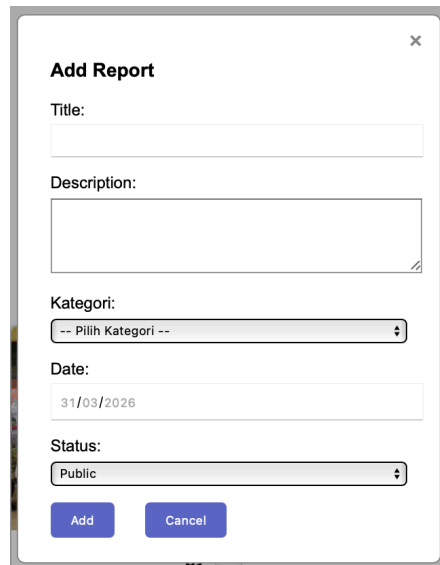
A modal dialog titled "Add Report" with a close button (X) in the top right corner. The form contains the following fields: "Title:" with a text input; "Description:" with a larger text area; "Kategori:" with a dropdown menu showing "-- Pilih Kategori --"; "Date:" with a date input showing "31/03/2026"; and "Status:" with a dropdown menu showing "Public". At the bottom are two buttons: "Add" and "Cancel".

Figure C11 : **Add Report** modal dialog

9. Manage Report Page

Create the Manage Report page (Figure C12). Requirements:

9.1 Report Information (Top Section)

- Load the title, description, category, date and status.
- Basic input validations: Title is required, Description is required, Date is required. Status is either public (0) or private (1).
- Display category name based on user **Category_Id**.
- Update button: Update the album. Redirect to Albums page.
- Reset button: Reset the form (load the original title and status).

Manage Report

Title :

Description :

Bidang ini menguji kepakaran peserta dalam membangunkan aplikasi web moden yang responsif dan dinamik. Fokus utama merangkumi reka bentuk antaramuka (UI/UX), pengaturcaraan front-end menggunakan HTML, CSS, dan JavaScript, serta pembangunan back-end melalui pengurusan pangkalan data dan

Kategori :

Date :

Status :

Report Photos





+

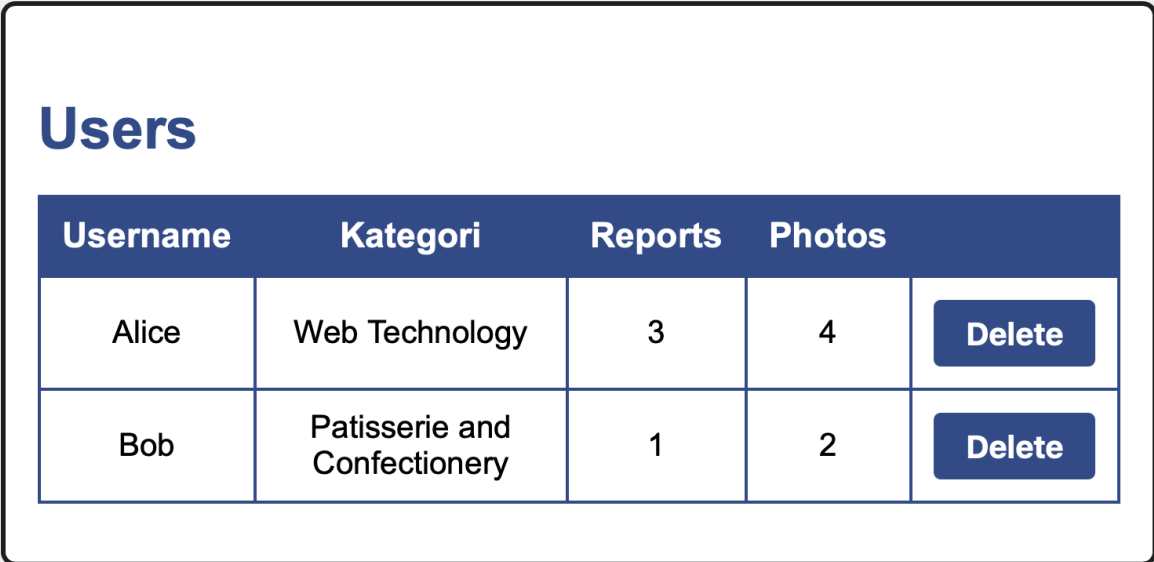
Figure C12 : **Manage Report** page

9.2 Report Photos (Bottom Section)

- List all photos in the album as thumbnails (visually cropped and resized to fit in square box).
- Clicking on a thumbnail will show the respective photo in full-screen mode.
- Delete (cross) button: Delete the photo.
- Add (plus) button: Open file picker. Select one or multiple photos and directly upload to server.
- Drag-and-drop: Drag-and-drop one or multiple photos and directly upload to server.
- Provide visual effect to highlight the valid drop region.
- Photos uploaded must be physically scaled down to fit within the 600 x 600 pixels boundary.
- (smaller photos no need to be resized) with original aspect ratio retained.
- Load the newly uploaded photos into the photo list.
- Use AJAX techniques for this to prevent page reload.

10. Users Page (Admin Only)

Create the Users page (**Figure C13**). Requirements:



The screenshot shows a web page titled "Users". Below the title is a table with the following structure:

Username	Kategori	Reports	Photos	
Alice	Web Technology	3	4	<button>Delete</button>
Bob	Patisserie and Confectionery	1	2	<button>Delete</button>

Figure C13 : Users page

- List all users with aggregate data (**username**, **albums count** and **photos count**):
 - **Reports count**: Total number of albums the user has.
 - **Photos count**: Total number of photos the user has.
- **Delete** button: Delete the user and all of the related records (albums, photos, etc.).

INSTRUCTIONS TO THE COMPETITOR

Save the files of your application in directory on the server and name it as "**CompetitorID_Module_C**", where **CompetitorID** is the IC number.

MATERIALS PROVIDED

ITEM	DESCRIPTION
Laravel 12 * Pre-downloaded projects via composer. Composer will not be installed and provided.	PHP framework https://laravel.com/docs/12.x
image folder	Folder containing images for testing purpose
Description	File containing the description of each competition category
categorys.csv	List of competition category

MARKING SCHEME SUMMARY

SECTION	CRITERION	MAX MARK
C1	Structure	2.00
C2	Database	5.25
C3	Authentication	7.50
C4	One Page Report Page	3.50
C5	Slide Page	6.50
C6	MyReport Page	7.25
C7	Report Information (Top Section)	3.50
C8	Report Photos (Bottom Section)	8.50
C9	Users page	3.00
C10	Design and layout	3.00
Total		50.00